



Inferences for Two Population Variances Transcript

Hello everyone, welcome to lesson 1 video 2. In this video, we will focus on inferences for two different population variances. This is important when we want to check whether two groups have similar variability or if one is more spread, the other one is not. The main topic in this video is F-test.

In our journey through higher day testing till now, our focus has been almost entirely on the mean. But in many real-world scenarios, the average isn't the most important metric. Sometimes we care much more about consistency, stability, risk or variability, right? Today, we are going to explore inferences of two population variances.

We will learn how to statistically compare the variability of two groups using a new tool which is called as F-statistics or F-distribution. Okay, so let's get started. So why do we actually want to compare variances? Let's consider a couple of examples.

We have example 1 which is from manufacturing, the example 2 which is from finance, right? So imagine two machines in a factory that both produce pistons, okay? We have two different machines from the same factory which produces pistons and we are trying to see whether the average diameter of the piston that they produce is exactly same or different, okay? So now that we are producing, we are trying to see whether the average diameter is same or not. We can definitely do a t-test to compare the diameters of the machines that they produced. But to test the consistency of the machines, sometimes we might want to look into the variability component or the variance component of it.

So let's say machine A's output is highly consistent while machine B's output is having high variability, right? For quality control, machine A is superior because it is more consistent, right? So how and what methods do we use for judging the consistency, judging the variability of the machine and so on, this is where the entire distribution which we call as F-test or F-distribution comes in, okay? Now let's say about the second example. Two investment firms might boost the same average annual return but if fund A is widely volatile, it has high variance, while fund B is stable and predictable, a risk-averse investor would always prefer the lower variance of fund B, right? So everyone would like to go for fund B because it doesn't have so much amount of volatility or variability, right? So in general, we are or we might not be always concerned about averages. There are situations when we might be keeping into focus or our focus might be only around some variability, right, or only about around the variance or what is more consistent, what is less consistent, what is more variant.

So all of those questions is what we are looking for, right? So this is a tool that we use. I told you about F-test or F-distribution. The tool for comparing variances is this F-test or F-distribution, okay? Now the Z and T-test were built to handle differences between the two population means but or let's say to test the population and its mean but F-



test is built to test the variances, okay? We need whenever we need to build variances between the two groups, we can easily compute their ratios or their ratio of the variances and that's what is an F-distribution, okay? So F-distribution is nothing but the ratio of two variances.

Let's say we have two groups and we have variances from the two groups when you compare them that's what the F-test is. It is described by two set of degrees of freedom. One, the numerators degrees of freedom which is the first variance and second the denominators degrees of freedom which is the second variance, okay? It is always non-negative because what we keep is whatever variance is higher we will keep that value at the top at the numerator, okay? That's by convention it is what we use because we want the F-value to be very very greater than 1 or let's say highly greater than 1 to understand that the first group is having more variability than the second group, okay? So once we take the ratio of the higher number versus the lower number, if they are comparable the value would be very close to 1, if they are not comparable if the numerator is greater than the denominator in that case F will be significantly greater than 1, right? Which means the first group is more variable or the numerator is more variable than the denominator, okay? So this is what we take into account, the test we use is called as F-test, we apply our familiar 4-step framework wherein we start by building the hypothesis, we then state our standard, then we test the statistic, then we get the result right.

So this is a 4-step process if you think of. Similarly, that's what we will follow in F-statistics. The null hypothesis is that the two population variances are same, so we start by saying that there is no differences, everything is same, that's what your null hypothesis is.

Remember I told you when you state null hypothesis it is mostly around equals to, greater than equals to, less than equals to, okay? Alternate hypothesis is not equals to, greater than, lesser than and so on, okay? So this is how you will overlay your hypothesis and then the third step is to test, right? So you will reduce the F-statistics which is nothing but the ratio of the variances S_1^2 square by S_2^2 square, the two sample variance that you will compute from the samples, you will take their ratios. If F value is very, very much greater than 1, this implies that the two groups are very much different in terms of their variability. If the value is comparable to 1, closer to 1, it means that it doesn't have a lot of variability but we will look into more details of it during the live lesson or live lecture and we will build around this intuition as well, okay? Just to help you understand, use an example.

Let's say you have ball bearings being produced by two machines and you have captured all of this data. Let's say machine 1, in machine 1 we have number of samples or number of ball bearings that we have captured is R21 and its variability was 0.025 and machine B, the number of ball bearings that we have sampled was 16 and their variances are 0.014. Now, our step 1 is to state the hypothesis which is both the



variances are same, alternatives they are not same, we will choose a level of significance which is our gold standard 5% or 10% or 1% whatever it is, then we will move into calculating the statistic which is our F statistics, the ratio between the two variances. Now, see that the value 1.79 is closer to, almost closer to 1, right? It doesn't look like very significant and that's why you see on this region, the p-value once we start looking into the area under the curve for the F distribution, we find that the p-value is not very very significant which is 0.25 which is not less than 0. Sorry, in this case, it should be less than 0.1 and hence we fail to reject null hypothesis.

All of these things is what we will cover in the live lesson. Now, I will help you understand how to actually draw area under the curve and find all of those probabilities because on PPTs, it becomes slightly difficult to make anyone understand that, right? So, in the live lesson, I will help you understand how p-values are calculated, how all of these things are calculated but at least from the intuition point of view, you understand that you start with step 1 which is stating the hypothesis, then you go move to step 2 which is setting the gold standard. Step 3 is when you will calculate your test statistic.

In this case, it is about variability, hence F distribution of F-test. You would compare both the variances and you have got 1.79 with degrees of freedom 20 and 15. This is because you have 21 observations, so $21-1$ is the degrees of freedom for numerator and you have 16 observations here, so $16-1$, 15 is the degrees of freedom for the denominator.

Now, you will calculate the p-value and now p-value is also calculated. First, you will find out the F-test value. Based on that F-test, we will find the p-value and then if the p-value is greater than your level of significance, that what we have learnt, right? It was 10%.

So, if the p-value is greater than level of significance, then you will say that we fail to reject null hypothesis. If it is lesser, we will say that the alternate hypothesis is true and that is what the conclusion, right? So, this is what we use and there are two different methods. One is the p-value method, the other is critical value method.

You can use any one of them and I will tell you about how to look into F-table and use critical value method during our live sessions, okay? So, this is again one other place wherein you can visualise F-distribution by number of degrees of freedom of the numerator, number of degrees of freedom of the denominator and the F-statistics. So, what happens when you see the F-statistics is closer to or lesser than 1, in that case, you see that the right tail p-value is getting changed, right? And what happens when your numerator increases or the degrees of freedom of the numerator increases and what happens when your degrees of freedom of the denominator increases? How does the entire F-distribution shape looks like? So, it's an interactive way of visualising how does the entire F-distribution gets shaped to using various different ways, let's



say, when your numerator increases like denominator is significantly constant or let's say it is constant, when your denominator increases while your numerator is constant, how does all of that affect and how does the F-statistic value helps you get to the p-value and the region says the rejections or acceptance regions which is what we'll learn in the live class, okay? I hope at least from the point of view of understanding when the question is about variability or when the question is about understanding are the two groups, among the two groups, which one is more variable, then by intuition, you should not have any doubt, the very first thing that would come to your mind is F-test or F-distribution, okay? That's what we have studied and this is a simple summary which I have just stated, so I hope all of these things are clear to everyone and let's meet in the next video. In summary, comparing variances allows us to test whether two populations differ in their spread, this is not only important in its own way but also is a prerequisite for techniques such as analysis of variance.

Next, we are going to study ANOVA or analysis of variance.

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